

Comparative Study of Classical CNN Architectures

ELCT918 - Lab Assignment 2 | LeNet-5, AlexNet and VGG16

Name: Amr Walid Elkamash

Student ID: 68-1941

Name: Mohamed Amr Mohamed Mostafa

Student ID: 68-1941

1. Objective and experimental setup

Three CNNs were trained from scratch on CIFAR-10 and CIFAR-100 to compare architecture, accuracy and training cost. AlexNet achieved the highest test accuracy under the shared 20-epoch budget.

Setting	Configuration
Framework and input	PyTorch; RGB images, 3 x 32 x 32
Data split per dataset	45,000 training; 5,000 validation; 10,000 test images
Preprocessing	ToTensor; pixel values scaled to [0, 1]; no augmentation or channel normalization
Training	20 epochs; batch size 64; Adam; learning rate 0.0001
Loss and regularization	CrossEntropyLoss; dropout 0.5 in AlexNet/VGG16 classifiers
Reproducibility and evaluation	Split seed 42; model seed 42 per run; final-epoch weights tested
Timing	Mean training time per epoch; excludes validation; GPU synchronization before/after training

The same settings were used for all models. A learning rate of 0.0001 was selected after VGG16 stalled at 0.001. The 20-epoch budget kept runtime practical.

2. Test results

Dataset	Model	Top-1 %	Top-5 %	Drop %	Time s/epoch
CIFAR10	LeNet5	48.41	-	51.59	7.65
CIFAR10	AlexNet	76.53	-	23.47	10.47
CIFAR10	VGG16	73.45	-	26.55	11.35
CIFAR100	LeNet5	17.43	42.79	82.57	7.71
CIFAR100	AlexNet	46.96	74.94	53.04	10.52
CIFAR100	VGG16	8.10	28.86	91.90	11.40

Accuracy drop (%) = 100 - test accuracy (%). Top-5 is reported only for CIFAR-100. Timings exclude validation; the GPU model was not recorded.

3. Implemented architectures

Output shapes are channels x height x width. $C = 10$ or 100 . K/S/P = kernel/stride/padding. Convolutions include biases; output layers produce logits.

LeNet-5 adaptation

Layer	Channels / units	K / S / P	Output
Input	3	-	$3 \times 32 \times 32$
Conv 1	3 to 6	5 / 1 / 0	$6 \times 28 \times 28$
Average pool 1	6	2 / 2 / 0	$6 \times 14 \times 14$
Conv 2	6 to 16	5 / 1 / 0	$16 \times 10 \times 10$
Average pool 2	16	2 / 2 / 0	$16 \times 5 \times 5$
Conv 3	16 to 120	5 / 1 / 0	$120 \times 1 \times 1$
Flatten	120	-	120
FC 1	120 to 84	-	84
FC output	84 to C	-	C

Tanh activation; RGB input and C outputs. Simplifications: ordinary average pooling, full channel connections and a linear output replace original trainable subsampling, partial connections and RBF output.

AlexNet adaptation

Layer	Channels / units	K / S / P	Output
Input	3	-	$3 \times 32 \times 32$
Conv 1	3 to 96	5 / 1 / 2	$96 \times 32 \times 32$
Max pool 1	96	2 / 2 / 0	$96 \times 16 \times 16$
Conv 2	96 to 256	5 / 1 / 2	$256 \times 16 \times 16$
Max pool 2	256	2 / 2 / 0	$256 \times 8 \times 8$
Conv 3	256 to 384	3 / 1 / 1	$384 \times 8 \times 8$
Conv 4	384 to 384	3 / 1 / 1	$384 \times 8 \times 8$
Conv 5	384 to 256	3 / 1 / 1	$256 \times 8 \times 8$
Max pool 3	256	2 / 2 / 0	$256 \times 4 \times 4$
Flatten	4096	-	4096
FC 1 + dropout	4096 to 4096	-	4096
FC 2 + dropout	4096 to 4096	-	4096
FC output	4096 to C	-	C

ReLU activation; smaller first kernel and pooling windows fit CIFAR inputs. Original response normalization and grouped connections are omitted. Dropout 0.5 follows hidden FC activations.

VGG16 adaptation

Layer	Channels / units	K / S / P	Output
Input	3	-	$3 \times 32 \times 32$
Conv 1.1	3 to 64	3 / 1 / 1	$64 \times 32 \times 32$
Conv 1.2	64 to 64	3 / 1 / 1	$64 \times 32 \times 32$
Max pool 1	64	2 / 2 / 0	$64 \times 16 \times 16$
Conv 2.1	64 to 128	3 / 1 / 1	$128 \times 16 \times 16$
Conv 2.2	128 to 128	3 / 1 / 1	$128 \times 16 \times 16$
Max pool 2	128	2 / 2 / 0	$128 \times 8 \times 8$
Conv 3.1	128 to 256	3 / 1 / 1	$256 \times 8 \times 8$
Conv 3.2	256 to 256	3 / 1 / 1	$256 \times 8 \times 8$
Conv 3.3	256 to 256	3 / 1 / 1	$256 \times 8 \times 8$
Max pool 3	256	2 / 2 / 0	$256 \times 4 \times 4$
Conv 4.1	256 to 512	3 / 1 / 1	$512 \times 4 \times 4$
Conv 4.2	512 to 512	3 / 1 / 1	$512 \times 4 \times 4$
Conv 4.3	512 to 512	3 / 1 / 1	$512 \times 4 \times 4$
Max pool 4	512	2 / 2 / 0	$512 \times 2 \times 2$
Conv 5.1	512 to 512	3 / 1 / 1	$512 \times 2 \times 2$
Conv 5.2	512 to 512	3 / 1 / 1	$512 \times 2 \times 2$
Conv 5.3	512 to 512	3 / 1 / 1	$512 \times 2 \times 2$
Max pool 5	512	2 / 2 / 0	$512 \times 1 \times 1$
Flatten	512	-	512
FC 1 + dropout	512 to 4096	-	4096
FC 2 + dropout	4096 to 4096	-	4096
FC output	4096 to C	-	C

ReLU after convolutions and hidden FC layers; dropout 0.5 after hidden FC activations. CIFAR input reduces the first FC input to 512. Output size is C ; no batch normalization.

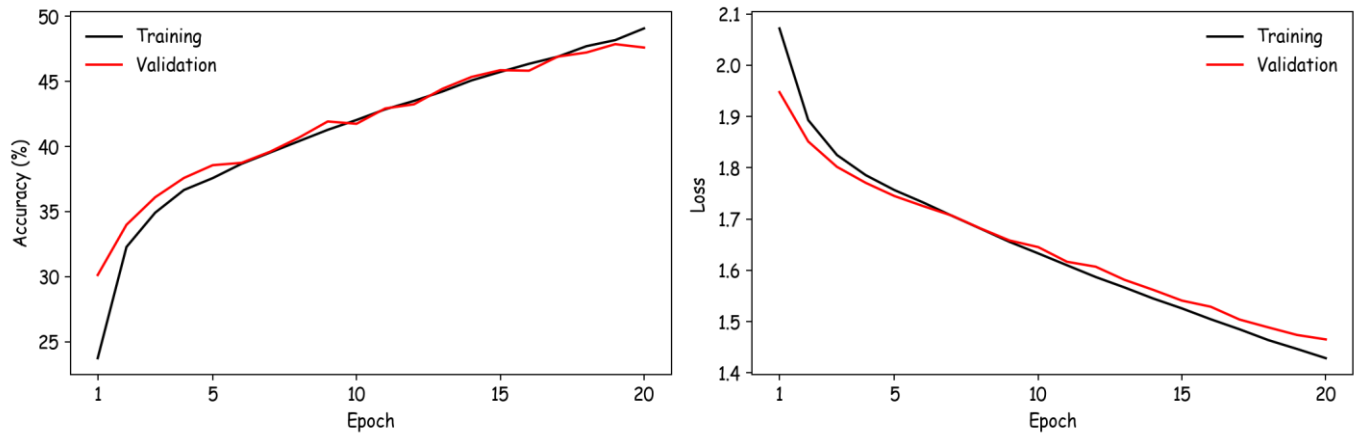
Depth and parameter comparison

Model	Conv + FC depth	CIFAR-10 parameters	CIFAR-100 parameters
LeNet-5	$3 + 2 = 5$	62,006	69,656
AlexNet	$5 + 3 = 8$	37,323,146	37,691,876
VGG16	$13 + 3 = 16$	33,638,218	34,006,948

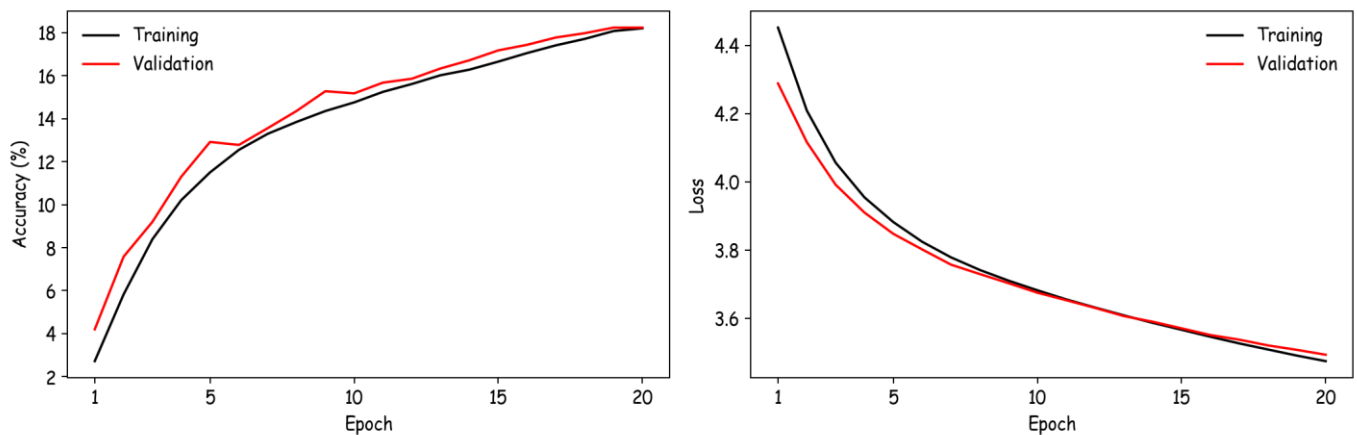
Depth counts convolutional and FC layers only. Including pooling, this LeNet variant has seven stages. Parameters include biases.

LeNet is compact; AlexNet is wider; VGG16 stacks small 3×3 filters. Two stride-1 3×3 layers cover a 5×5 receptive field, using $18M^2$ versus $25M^2$ kernel weights at equal width M . AlexNet has more parameters here because its first FC input is 4096 rather than 512.

4. LeNet-5 learning behavior



LeNet5 on CIFAR10: accuracy (left) and loss (right).



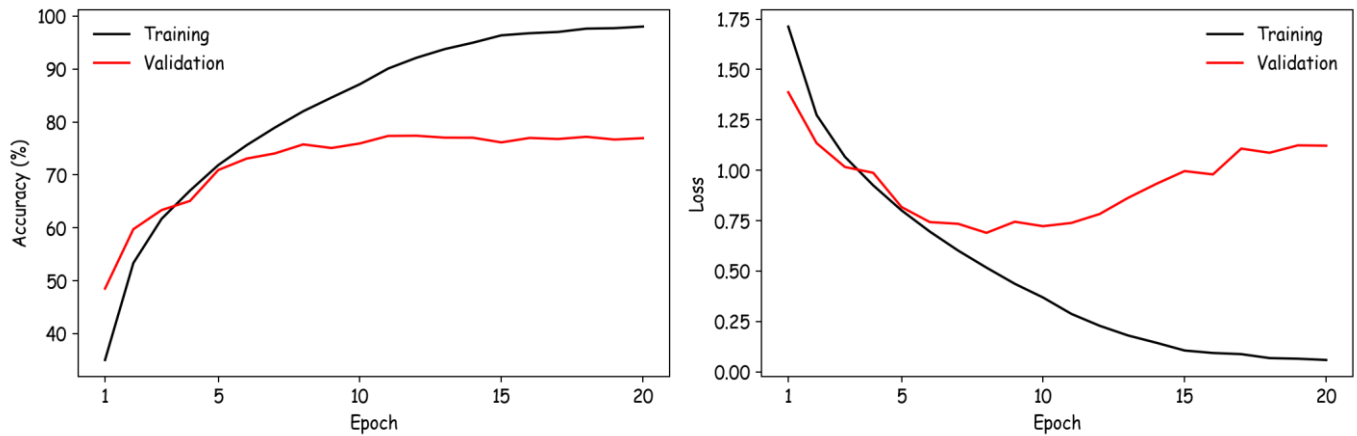
LeNet5 on CIFAR100: accuracy (left) and loss (right).

LeNet ends at 49.07% training / 47.60% validation accuracy on CIFAR-10 and 18.21% / 18.24% on CIFAR-100. Small gaps and declining losses show little overfitting. Low accuracy suggests limited fit under this setup.

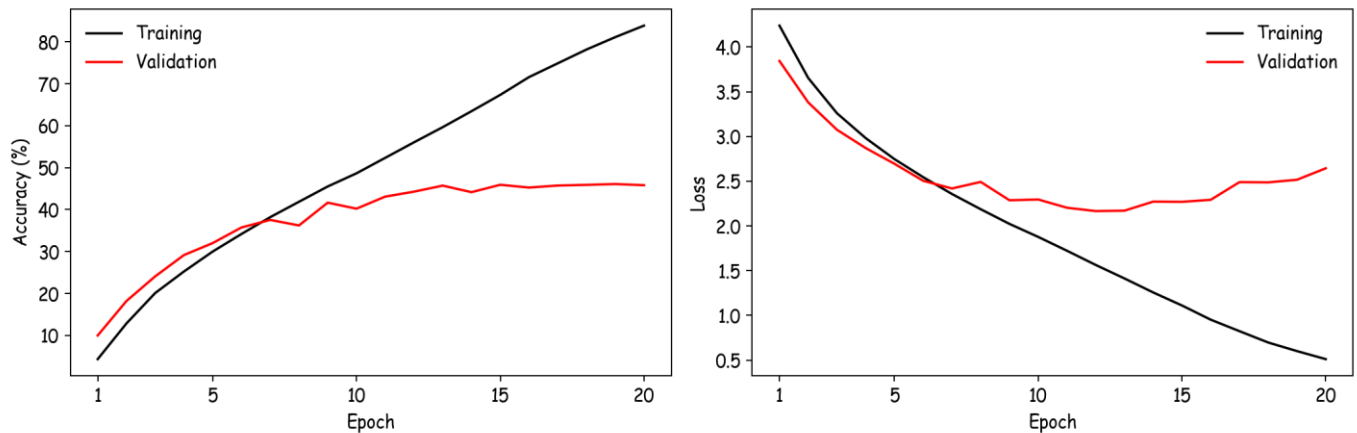
Why CIFAR-100 is harder

CIFAR-100 has ten times more classes but the same total development set: 500 images per class versus 5,000 for CIFAR-10 before splitting. Less data per class and finer distinctions increase difficulty.

5. AlexNet learning behavior



AlexNet on CIFAR10: accuracy (left) and loss (right).



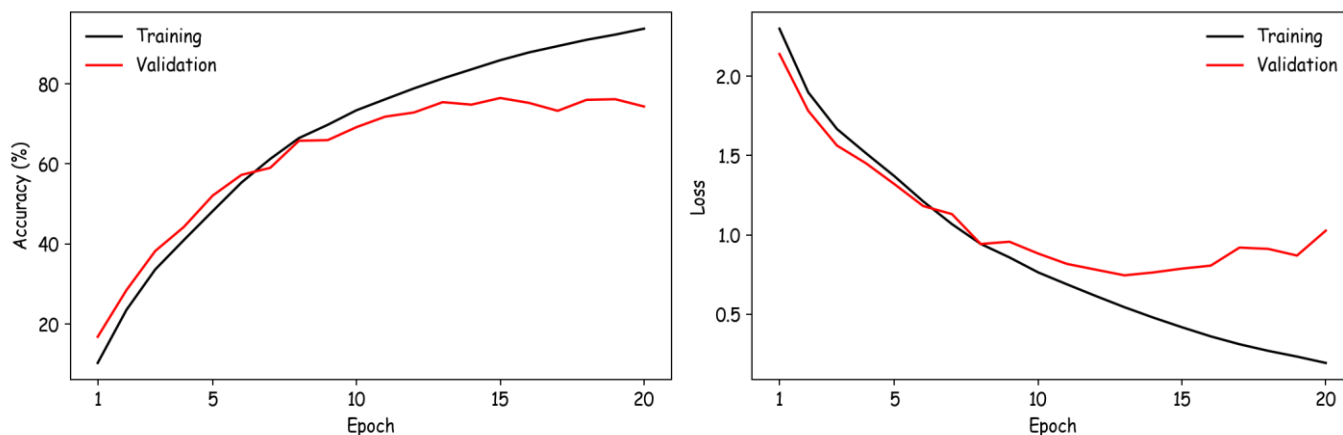
AlexNet on CIFAR100: accuracy (left) and loss (right).

AlexNet ends at 98.02% training / 76.94% validation accuracy on CIFAR-10 and 83.88% / 45.82% on CIFAR-100. Rising validation loss alongside falling training loss shows clear overfitting. Test results use final-epoch weights.

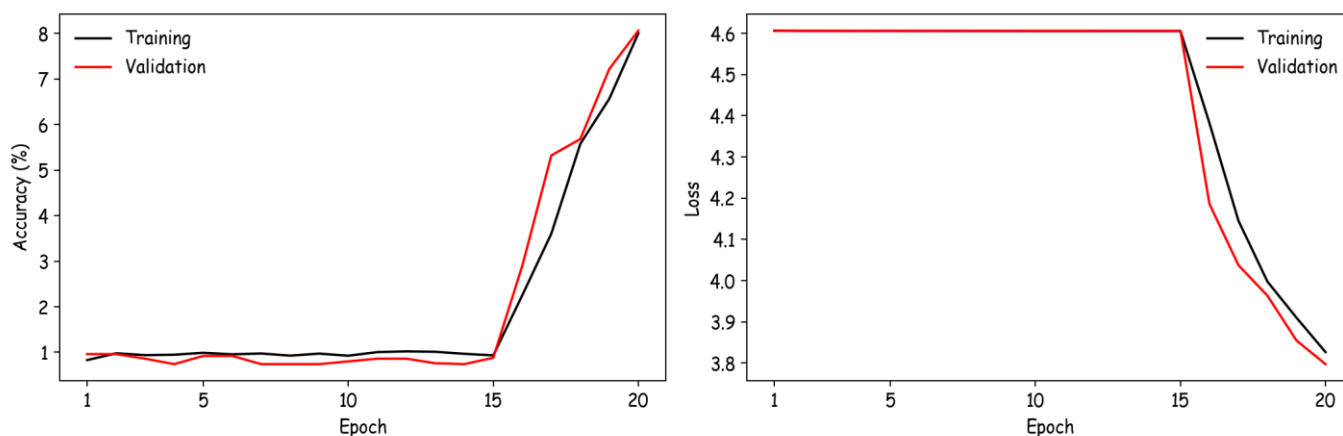
Accuracy versus computational cost

AlexNet exceeds VGG16 by 3.08 accuracy points on CIFAR-10 and 38.86 on CIFAR-100, while training faster per epoch. Extra depth gives no measured benefit here. Timing depends on hardware, computation and data loading, not parameter count alone.

6. VGG16 learning behavior



VGG16 on CIFAR10: accuracy (left) and loss (right).



VGG16 on CIFAR100: accuracy (left) and loss (right).

VGG16 overfits CIFAR-10: final training accuracy is 93.75% versus 74.32% validation. On CIFAR-100, learning starts only after about 15 epochs; final accuracies are 8.00% / 8.06%. Both curves still improve, indicating incomplete convergence rather than overfitting.

7. Conclusion and limitations

AlexNet gives the best test accuracy and the largest training-validation gaps. LeNet is smallest and fastest. VGG16 overfits CIFAR-10 but learns too late on CIFAR-100. These findings apply to one seed, simplified models and a fixed budget, not an intrinsic ranking of the original networks.

GitHub repository URL: <https://github.com/TwingleMora/AI-Accelerators-Course>